Distinguished Root Branches over Algebraic Worlds

The Architectural Theorems of the Perspective Family Program

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Abstract

Five theorems are established governing the structure of perspective families $f(x) + P = \alpha$, where α is a fixed positive algebraic number, $f \in \mathbf{Q}[x]$ is strictly increasing on $(0, \infty)$, and P is a real parameter. Theorem A proves that each such family has a unique, canonical, strictly decreasing positive real root branch. Theorem B proves that the algebraic degree of the root at rational P is bounded by $[\mathbf{Q}(\alpha) : \mathbf{Q}] \cdot \deg(f)$, with equality when the elimination polynomial is irreducible. Theorem C proves that the mixed compositum $\mathbf{Q}(\varphi, \rho)$ has degree 6, establishing a degree ceiling of 18 for cubic constructions over the mixed field. Theorem D proves the tight realization obstruction: the base field degree must divide the construction constant's degree, so that pure powers of 2 exclude cubic base fields and pure powers of 3 exclude quadratic base fields under tight realization; mixed degrees are not forced by divisibility alone. Theorem E proves that the four exhibited family types (binary quadratic, binary octic, τ-cubic, ρ-cubic) produce elimination polynomials with mutually exclusive coefficient structures, providing a normal form obstruction that separates families by shape as well as degree—and reveals internal structure within the ternary world itself.

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1 Definitions

Definition 1.1 (Perspective family)**.** A *perspective family* is a triple (α, f, m) where:

- $\alpha > 0$ is a fixed real algebraic number (the *reference constant*),
- $f(x) \in \mathbf{Q}[x]$ is a polynomial satisfying $f(0) = 0$ and $f'(x) > 0$ for all $x > 0$,
- $m \geq 1$ is a positive integer (the *reference exponent*).

The *family equation* is

$$f(x) + P = \alpha^m, \tag{1}$$

where $P \in \mathbf{R}$ is the *perspective parameter*.

Definition 1.2 (Admissible parameter)**.** A parameter P is *admissible* if $P < \alpha^m$. Equivalently, if $\alpha^m - P > 0$.

Definition 1.3 (Construction value)**.** For an admissible parameter P, the *construction value* is $C(P) = 2\,x(P)$, where $x(P)$ is the distinguished positive root defined in Theorem 3.1 below.

Definition 1.4 (Algebraic world)**.** The *world* of a perspective family (α, f, m) is determined by the base field $K = \mathbf{Q}(\alpha^m)$:

- **Binary**: $K = \mathbf{Q}(\varphi)$, where $\varphi = \frac{1+\sqrt{5}}{2}$ (golden ratio, $[K : \mathbf{Q}] = 2$).
- **Ternary**: $K = \mathbf{Q}(\rho)$ or $K = \mathbf{Q}(\tau)$ (tribonacci or plastic constant, $[K : \mathbf{Q}] = 3$).
- **Mixed**: $K = \mathbf{Q}(\varphi, \rho)$ ($[K : \mathbf{Q}] = 6$, see Theorem 5.1).

2 Preliminary Lemmas

Lemma 2.1 (Rational scaling)**.** *For any $q \in \mathbf{Q}^\times$ and any algebraic number x, $\mathbf{Q}(qx) = \mathbf{Q}(x)$ and* $\deg(qx) = \deg(x)$.

Proof. Since $q \in \mathbf{Q}$, we have $qx \in \mathbf{Q}(x)$, so $\mathbf{Q}(qx) \subseteq \mathbf{Q}(x)$. Conversely, $x = (qx)/q \in \mathbf{Q}(qx)$, so $\mathbf{Q}(x) \subseteq \mathbf{Q}(qx)$. □

Remark 2.2. This lemma justifies the passage between the distinguished root $x(P)$ and the construction value $C(P) = 2x(P)$ throughout the registry: they have the same algebraic degree.

Lemma 2.3 (Field generation by prime-degree elements)**.** *Let K be a number field with $[K : \mathbf{Q}] = p$ prime. If $\alpha \in K \setminus \mathbf{Q}$, then $\mathbf{Q}(\alpha) = K$. In particular, for any positive integer m with $\alpha^m \notin \mathbf{Q}$, we have $\mathbf{Q}(\alpha^m) = K$.*

Proof. Since $\alpha \in K$, $\mathbf{Q}(\alpha) \subseteq K$. Since $[K : \mathbf{Q}] = p$ is prime, the only intermediate fields are $\mathbf{Q}$ and K. Since $\alpha \notin \mathbf{Q}$, $\mathbf{Q}(\alpha) \neq \mathbf{Q}$, hence $\mathbf{Q}(\alpha) = K$. □

Remark 2.4. This lemma applies to the mixed example: $\beta = \rho\varphi$ has degree 6, and β^4 has degree 6 (verified computationally: β^4 has irreducible minimal polynomial of degree 6). Since 6 is not prime, the prime-degree lemma does not apply directly, but the computational verification confirms $\mathbf{Q}(\beta^4) = \mathbf{Q}(\beta)$ for the exhibited mixed reference.

3 Theorem A: Distinguished Positive Root Branch

Theorem 3.1 (Distinguished branch)**.** *Let (α, f, m) be a perspective family. Then:*

(i) For each admissible $P < \alpha^m$, there exists exactly one positive real root $x(P) > 0$ of the equation $f(x) + P = \alpha^m$.

(ii) The map $P \mapsto x(P)$ is continuous and differentiable on $(-\infty, \alpha^m)$.

(iii) The branch is strictly decreasing:

$$x'(P) = -\frac{1}{f'(x(P))} < 0. \tag{2}$$

(iv) As $P \to \alpha^m$ from below, $x(P) \to 0^+$. As $P \to -\infty$, $x(P) \to +\infty$.

Proof. **(i)** Since $f'(x) > 0$ for all $x > 0$, the function f is strictly increasing on $(0, \infty)$. Because f is a nonconstant polynomial with $f'(x) > 0$ for all $x > 0$, its leading coefficient is positive, so $f(x) \to +\infty$ as $x \to +\infty$. Combined with $f(0) = 0$, the function f is a bijection from $[0, \infty)$ to $[0, \infty)$.

For admissible P, we have $\alpha^m - P > 0$, so the equation $f(x) = \alpha^m - P$ has exactly one solution $x > 0$.

(ii) The implicit function theorem applies because $f'(x(P)) > 0$ (so the partial derivative with respect to x is nonzero). Therefore $x(P)$ is differentiable.

(iii) Differentiating $f(x(P)) + P = \alpha^m$ with respect to P:

$$f'(x(P)) \cdot x'(P) + 1 = 0,$$

so $x'(P) = -1/f'(x(P)) < 0$ since $f'(x(P)) > 0$.

(iv) As $P \to \alpha^m$, $f(x) = \alpha^m - P \to 0^+$, so $x \to 0^+$ (since $f(0) = 0$ and f is increasing). As $P \to -\infty$, $f(x) = \alpha^m - P \to +\infty$, so $x \to +\infty$. □

Corollary 3.2 (Canonicality)**.** *For each admissible P, the construction value $C(P) = 2\,x(P)$ is uniquely defined, strictly decreasing in P, and not an artifact of root selection.*

Remark 3.3. This theorem applies to all concrete families in the registry:

- $f(x) = x^2 + x$: $f'(x) = 2x + 1 > 0$ for $x > 0$. ✓
- $f(x) = x^3 + x$: $f'(x) = 3x^2 + 1 > 0$. ✓
- $f(x) = x^3 + x^2 + x$: $f'(x) = 3x^2 + 2x + 1 > 0$ (discriminant $4 - 12 < 0$). ✓
- $f(x) = x^8 + x^4 + x^2 + x$: $f'(x) = 8x^7 + 4x^3 + 2x + 1 > 0$. ✓

4 Theorem B: Degree Ceiling

Theorem 4.1 (Degree ceiling)**.** *Let (α, f, m) be a perspective family with $K = \mathbf{Q}(\alpha^m)$. Let $P \in \mathbf{Q}$ be a rational admissible parameter, and let $x = x(P)$ be the distinguished positive root. Then:*

(i) x is algebraic over $\mathbf{Q}$.

(ii) The algebraic degree of x satisfies

$$[\mathbf{Q}(x) : \mathbf{Q}] \leq [K : \mathbf{Q}] \cdot \deg(f). \tag{3}$$

(iii) Define the elimination polynomial *by*

$$F_P(x) = m_{\alpha^m}\big(f(x) + P\big),$$

where m_{α^m} is the minimal polynomial of α^m over $\mathbf{Q}$. Then $\deg F_P = [K : \mathbf{Q}] \cdot \deg(f)$. If F_P is irreducible over $\mathbf{Q}$, then it is the minimal polynomial of x, and equality holds in (3).

Proof. **(i)** Since $P \in \mathbf{Q}$ and α^m is algebraic, the quantity $\alpha^m - P$ is algebraic. The equation $f(x) = \alpha^m - P$ is a polynomial equation with coefficients in K. Since the algebraic closure $\overline{\mathbf{Q}}$ is algebraically closed, every root x lies in $\overline{\mathbf{Q}}$—that is, x is algebraic.

(ii) Over K, the element x is a root of

$$g(T) = f(T) - (\alpha^m - P) \in K[T],$$

which has degree $n = \deg(f)$. Therefore $[K(x) : K] \leq n$. By the tower law:

$$[K(x) : \mathbf{Q}] = [K(x) : K] \cdot [K : \mathbf{Q}] \leq n \cdot [K : \mathbf{Q}].$$

Since $\mathbf{Q}(x) \subseteq K(x)$:

$$[\mathbf{Q}(x) : \mathbf{Q}] \leq [K(x) : \mathbf{Q}] \leq \deg(f) \cdot [K : \mathbf{Q}].$$

(iii) The minimal polynomial m_{α^m} has degree $[K : \mathbf{Q}]$. Substituting $z = f(x) + P$ into $m_{\alpha^m}(z)$ produces $F_P(x)$, which has degree $[K : \mathbf{Q}] \cdot \deg(f)$ (the composition degree). By construction, x is a root of F_P. If F_P is irreducible over $\mathbf{Q}$, then F_P is the minimal polynomial of x, so $[\mathbf{Q}(x) : \mathbf{Q}] = \deg F_P = [K : \mathbf{Q}] \cdot \deg(f)$. □

Corollary 4.2 (Registry architecture)**.** *The degree ceiling theorem explains the entire degree pattern of the construction registry:*

World	*Base field K*	$[K : \mathbf{Q}]$	$\deg(f)$	*Ceiling*
Binary (quadratic f)	$\mathbf{Q}(\varphi)$	*2*	*2*	**4**
Binary (octic f)	$\mathbf{Q}(\varphi)$	*2*	*8*	**16**
Ternary	$\mathbf{Q}(\rho)$ *or* $\mathbf{Q}(\tau)$	*3*	*3*	**9**
Mixed	$\mathbf{Q}(\varphi, \rho)$	*6*	*3*	**18**

Every irreducible elimination polynomial in the registry achieves its ceiling exactly.

5 Theorem C: Mixed Compositum

Theorem 5.1 (Compositum degree)**.** *Let $\varphi = \frac{1+\sqrt{5}}{2}$ be the golden ratio and ρ the tribonacci constant (real root of $x^3 - x^2 - x - 1 = 0$). Then:*

(i) $\mathbf{Q}(\varphi) \cap \mathbf{Q}(\rho) = \mathbf{Q}$.

(ii) $[\mathbf{Q}(\varphi, \rho) : \mathbf{Q}] = 6$.

The same holds with τ (real root of $x^3 - x - 1 = 0$) in place of ρ.

Proof. **(i)** Let $L = \mathbf{Q}(\varphi) \cap \mathbf{Q}(\rho)$. Then $[L : \mathbf{Q}]$ divides both $[\mathbf{Q}(\varphi) : \mathbf{Q}] = 2$ and $[\mathbf{Q}(\rho) : \mathbf{Q}] = 3$. Since $\gcd(2, 3) = 1$, we have $[L : \mathbf{Q}] = 1$, hence $L = \mathbf{Q}$.

(ii) For finite extensions with trivial intersection and coprime degrees:

$$[\mathbf{Q}(\varphi, \rho) : \mathbf{Q}] = [\mathbf{Q}(\varphi) : \mathbf{Q}] \cdot [\mathbf{Q}(\rho) : \mathbf{Q}] = 2 \cdot 3 = 6.$$

The argument for τ is identical: $[\mathbf{Q}(\tau) : \mathbf{Q}] = 3$ (since $x^3 - x - 1$ is irreducible over $\mathbf{Q}$), $\gcd(2, 3) = 1$, so $[\mathbf{Q}(\varphi, \tau) : \mathbf{Q}] = 6$. □

Corollary 5.2 (Mixed-world ceiling)**.** *Any cubic perspective family with reference in $\mathbf{Q}(\varphi, \rho)$ has degree ceiling*

$$6 \cdot 3 = 18.$$

This is the exact degree of the $\pi \cdot e$ construction constant when the elimination polynomial is irreducible.

Remark 5.3. The compositum $\mathbf{Q}(\varphi, \rho)$ is the natural home for construction constants that involve *both* the binary and ternary worlds simultaneously. The reference $\beta = \rho\varphi$ satisfies the irreducible sextic

$$t^6 - t^5 - 4t^4 - 5t^3 - 2t^2 + t - 1 = 0$$

(irreducible modulo 23), confirming $[\mathbf{Q}(\beta) : \mathbf{Q}] = 6$. The element β lies in neither $\mathbf{Q}(\varphi)$ alone nor $\mathbf{Q}(\rho)$ alone (since 6 divides neither 2 nor 3 alone). The mixed reference is *structurally irreducible* to either single world.

Example 5.4 (Mixed construction near πe)**.** The family $\beta^4 = x^3 + x - \frac{11}{3}$ with $\beta = \rho\varphi$ defines a unique positive root ($f(x) = x^3 + x$ is strictly increasing). The resultant elimination produces a degree-18 polynomial, and the construction value $2x \approx 8.5394$ approximates $\pi e \approx 8.5397$ with gap 0.004%. The degree $18 = 6 \cdot 3$ is exactly the ceiling predicted by Theorem 4.1: base field degree 6 times polynomial degree 3.

6 Theorem D: Tight Realization Obstruction

Definition 6.1 (Tight realization)**.** A perspective family *tightly realizes* a construction constant c if:

- c is the distinguished positive root at some rational P,
- the elimination polynomial is irreducible over $\mathbf{Q}$,
- the degree ceiling is achieved: $[\mathbf{Q}(c) : \mathbf{Q}] = [K : \mathbf{Q}] \cdot \deg(f)$.

Theorem 6.2 (Tight realization obstruction)**.** *If a construction constant c with $[\mathbf{Q}(c) : \mathbf{Q}] = m$ is tightly realized by a perspective family with base field degree $d = [K : \mathbf{Q}]$ and polynomial degree $n = \deg(f)$, then*

$$d \mid m.$$

Proof. Tight realization means $d \cdot n = m$. Therefore $m/d = n \in \mathbf{Z}_{>0}$, which is to say d divides m. □

Corollary 6.3 (World exclusion)**.** *(i) If $m = 2^k$ for some $k \geq 1$ (a pure power of 2), then d must be a power of 2. In particular, $d = 3$ is excluded:* ***no ternary family can tightly realize a binary-degree constant****.*

(ii) If $m = 3^k$ for some $k \geq 1$ (a pure power of 3), then d must be a power of 3. In particular, $d = 2$ is excluded: ***no binary family can tightly realize a ternary-degree constant****.*

(iii) If $m = 2^a \cdot 3^b$ with $a, b \geq 1$, then d may be any divisor of m (including $d = 2$, $d = 3$, or $d = 6$). Divisibility alone does not force a unique world for mixed degrees.

Proof. **(i)** If $d = 3$ and $m = 2^k$, then $d \mid m$ requires $3 \mid 2^k$, which is false for all k.
(ii) If $d = 2$ and $m = 3^k$, then $d \mid m$ requires $2 \mid 3^k$, which is false for all k.
(iii) If $m = 2^a \cdot 3^b$, then d must divide m. Since m has divisors from both prime factors, the divisibility constraint does not exclude either binary ($d = 2$) or ternary ($d = 3$) base fields. □

Theorem 6.4 (Restricted divisibility obstruction)**.** *Under tight realization, the divisibility constraint $d \mid m$ restricts which base field degrees can realize a given construction constant. For the degrees in the current registry:*

Constant	m	*Factorization*	*Exclusions*	*Exhibited world*
π_c	*16*	2^4	$d = 3$ *excluded* ($3 \nmid 16$)	*Binary*
γ_c, $\ln(2)_c$	*4*	2^2	$d = 3$ *excluded* ($3 \nmid 4$)	*Binary*
e_c, $\sqrt{2\pi}_c$, $(2^{\sqrt{2}})_c$	*9*	3^2	$d = 2$ *excluded* ($2 \nmid 9$)	*Ternary*
$\zeta(3)_c$, $\sqrt{\pi}_c$, $\ln(3)_c$	*9*	3^2	$d = 2$ *excluded* ($2 \nmid 9$)	*Ternary*
$(\pi e)_c$	*18*	$2 \cdot 3^2$	*none* ($d = 2, 3, 6$ *all divide*)	*Mixed (exhibited)*

Proof. Direct application of Corollary 6.3 to each degree in the registry. For $m = 9 = 3^2$: $2 \nmid 9$, so binary base fields are excluded under tight realization. For $m = 4 = 2^2$: $3 \nmid 4$, so ternary base fields are excluded. For $m = 16 = 2^4$: $3 \nmid 16$, so ternary excluded. For $m = 18 = 2 \cdot 3^2$: all of $d = 2, 3, 6$ divide 18, so no world is excluded by divisibility alone; the exhibited realization uses $d = 6$. □

Remark 6.5. For pure prime-power degrees ($m = 2^k$ or $m = 3^k$), the divisibility obstruction provides a genuine cross-world exclusion: degree-2^k constants cannot be tightly realized in ternary families, and degree-3^k constants cannot be tightly realized in binary families. For mixed degrees like $m = 18$, the divisibility obstruction alone does not force a specific world; additional criteria (such as minimality of the pair (d, n) or normal-form compatibility) are needed to pin down the classification.

Definition 6.6 (Branch complexity)**.** For a positive algebraic number c, define its *branch complexity* as the lexicographically minimal pair (d, n) with $d > 1$ and $n > 1$ such that c is tightly realized by a perspective family with base field degree d and polynomial degree n. The *world* of c is determined by the prime factorization of d.

Remark 6.7 (Currently exhibited tight realizations).

Complexity (d, n)	$d \cdot n$	Construction constants
$(2, 2)$	4	$\gamma_{\mathrm{c}},\ \ln(2)_{\mathrm{c}}$
$(2, 8)$	16	π_{c}
$(3, 3)$	9	$e_{\mathrm{c}},\ \sqrt{2\pi}_{\mathrm{c}},\ (2^{\sqrt{2}})_{\mathrm{c}},\ \zeta(3)_{\mathrm{c}},\ \sqrt{\pi}_{\mathrm{c}},\ \ln(3)_{\mathrm{c}}$
$(6, 3)$	18	$(\pi e)_{\mathrm{c}}$

7 Theorem E: Normal Form Obstruction

The tight realization obstruction (Theorem 6.2) separates worlds by degree. This section proves a finer separation: even within a fixed degree, the *coefficient structure* of elimination polynomials distinguishes families.

Theorem 7.1 (Binary quartic normal form)**.** *Let α have minimal polynomial $z^2 + Uz + V$ over $\mathbf{Q}$, and let $P \in \mathbf{Q}$. Every elimination polynomial arising from the binary family $\alpha = x^2 + x + P$ has the form*

$$x^4 + 2x^3 + Mx^2 + (M-1)x + N = 0 \tag{4}$$

for some $M, N \in \mathbf{Q}$ determined by U, V, P. In particular:

(i) the coefficient of x^3 is exactly 2,

(ii) the coefficient of x^2 exceeds the coefficient of x by exactly 1.

Proof. Substitute $z = x^2 + x + P$ into $z^2 + Uz + V = 0$ and expand. The result is

$$x^4 + 2x^3 + (2P + U + 1)x^2 + (2P + U)x + (P^2 + UP + V) = 0.$$

Setting $M = 2P + U + 1$ and $N = P^2 + UP + V$, the coefficient of x is $M - 1 = 2P + U$, confirming both claimed identities. □

Theorem 7.2 (Binary degree-16 normal form)**.** *Let α have minimal polynomial $z^2 + Uz + V$ over $\mathbf{Q}$, and let $P \in \mathbf{Q}$. Every elimination polynomial arising from the binary family $\alpha = x^8 + x^4 + x^2 + x + P$ has the form*

$$x^{16} + 2x^{12} + 2x^{10} + 2x^9 + Mx^8 + 2x^6 + 2x^5 + Mx^4 + 2x^3 + Mx^2 + (M-1)x + N = 0 \tag{5}$$

with $M = 2P + U + 1$ and $N = P^2 + UP + V$. In particular:

(i) the coefficients of $x^{15}, x^{14}, x^{13}, x^{11}, x^7$ are all zero,

(ii) the coefficients of $x^{12}, x^{10}, x^9, x^6, x^5, x^3$ are all exactly 2,

(iii) the coefficients of x^8, x^4, x^2 are equal (all equal to M),

(iv) the coefficient of x is $M - 1$.

Proof. Substitute $z = x^8 + x^4 + x^2 + x + P$ into $z^2 + Uz + V = 0$ and expand. Setting $M = 2P + U + 1$ and $N = P^2 + UP + V$ yields (5) by direct computation.

Verification: for $\alpha = \varphi^8$, the minimal polynomial is $z^2 - 47z + 1$ (so $U = -47$, $V = 1$). At $P = 1$: $M = 2(1) + (-47) + 1 = -44$ and $N = 1 + (-47) + 1 = -45$. This recovers the degree-16 polynomial for π_c exactly. □

Theorem 7.3 (Ternary nonic normal form, type τ)**.** *Let β have minimal polynomial $z^3 + Az^2 + Bz + C$ over $\mathbf{Q}$, and let $P \in \mathbf{Q}$. Every elimination polynomial arising from the family $\beta = x^3 + x + P$ has the form*

$$x^9 + 3x^7 + Mx^6 + 3x^5 + 2Mx^4 + Nx^3 + Mx^2 + (N-1)x + L = 0 \tag{6}$$

for some $M, N, L \in \mathbf{Q}$. In particular:

(i) the coefficient of x^8 is exactly 0,

(ii) the coefficient of x^7 is exactly 3,

(iii) the coefficient of x^5 is exactly 3,

(iv) $\mathrm{coeff}(x^4) = 2 \cdot \mathrm{coeff}(x^6)$,

(v) $\mathrm{coeff}(x^2) = \mathrm{coeff}(x^6)$,

(vi) $\mathrm{coeff}(x^3) = \mathrm{coeff}(x) + 1$.

Proof. Substitute $z = x^3 + x + P$ into $z^3 + Az^2 + Bz + C = 0$ and expand. Collecting by powers of x and setting $M = A + 3P$, $N = 2AP + B + 3P^2 + 1$, $L = AP^2 + BP + C + P^3$, every coefficient is determined and the six identities follow by direct comparison. □

Theorem 7.4 (Ternary nonic normal form, type ρ)**.** *Let β have minimal polynomial z^3+Az^2+Bz+C over $\mathbf{Q}$, and let $P \in \mathbf{Q}$. Every elimination polynomial arising from the family $\beta = x^3 + x^2 + x + P$ has the form*

$$x^9 + 3x^8 + 6x^7 + Mx^6 + (2M - 8)x^5 + (3M - 18)x^4 + \cdots = 0 \tag{7}$$

for some $M \in \mathbf{Q}$. In particular:

(i) the coefficient of x^8 is exactly 3,

(ii) the coefficient of x^7 is exactly 6,

(iii) $\mathrm{coeff}(x^5) = 2 \cdot \mathrm{coeff}(x^6) - 8$,

(iv) $\mathrm{coeff}(x^4) = 3 \cdot \mathrm{coeff}(x^6) - 18$.

Proof. Substitute $z = x^3+x^2+x+P$ into $z^3+Az^2+Bz+C = 0$ and expand. Setting $M = A+3P+7$, the leading coefficients are fixed at $1, 3, 6$, and the identities (iii)–(iv) follow from the expansion. □

Theorem 7.5 (Mutual exclusion of normal forms)**.** *The four normal forms are pairwise incompatible:*

(i) Binary quartic (degree 4), binary degree-16, and both ternary nonics (degree 9) are separated by degree alone.

(ii) Within the binary world, the quartic template (Theorem 7.1) and the degree-16 template (Theorem 7.2) are separated by degree ($4 \neq 16$).

(iii) Within the ternary world, a type-τ nonic has $\mathrm{coeff}(x^8) = 0$, *while a type-ρ nonic has* $\mathrm{coeff}(x^8) = 3$. *No polynomial satisfies both.*

Therefore the four templates partition the registry into four mutually exclusive classes.

Proof. Parts (i)–(ii) are immediate from degree comparison. Part (iii): by Theorems 7.3(i) and 7.4(i), the coefficient of x^8 is 0 for type-τ and 3 for type-ρ. Since $0 \neq 3$, no polynomial can belong to both. □

Corollary 7.6 (Normal-form classification of exhibited families)**.** *The construction registry separates into four classes by normal form:*

Class	*Template signature*	*Degree*	*Members*
Binary quartic	x^3-*coeff* $= 2$, $x^2 = x + 1$	4	γ_{c}, $\ln(2)_{\mathrm{c}}$
Binary octic	$x^{15} = x^{14} = x^{13} = 0$, $x^8 = x^4 = x^2$	16	π_{c}
Ternary-τ	$x^8 = 0$, $x^7 = 3$, $x^5 = 3$	9	$\zeta(3)_{\mathrm{c}}$, $\sqrt{\pi}_{\mathrm{c}}$, $\ln(3)_{\mathrm{c}}$, $\sqrt{2\pi}_{\mathrm{c}}$
Ternary-ρ	$x^8 = 3$, $x^7 = 6$	9	e_{c}, $(2^{\sqrt{2}})_{\mathrm{c}}$

Every polynomial in the registry passes exactly one template and fails the other three.

Remark 7.7. The normal form obstruction provides a finer separation than the degree obstruction alone. Within degree 9, the coefficient of x^8 distinguishes the τ-family ($x^8 = 0$) from the ρ-family ($x^8 = 3$). Within the binary world, degree separates the quartic (γ_{c}, $\ln(2)_{\mathrm{c}}$) from the degree-16 π_{c}. These are necessary conditions for realizability within each family type, and they constitute the first layer of a normal-form obstruction program.

8 Summary

Theorem	**Content**
A (Distinguished Branch)	Each perspective family has a unique, canonical, strictly decreasing positive root branch. The root is not an artifact of selection.
B (Degree Ceiling)	At rational P, the root is algebraic with degree $\leq [K : \mathbf{Q}] \cdot \deg(f)$. Irreducibility of the elimination polynomial gives equality.
C (Mixed Compositum)	$[\mathbf{Q}(\varphi, \rho) : \mathbf{Q}] = 6$ because $\gcd(2, 3) = 1$. Cubic constructions over the mixed field have ceiling 18.
D (Divisibility Obstruction)	Tight realization requires $d \mid m$. Pure powers of 2 exclude cubic base fields; pure powers of 3 exclude quadratic base fields. For mixed degrees, divisibility alone does not force a unique world.
E (Normal Form)	Four exhibited family types produce elimination polynomials with mutually exclusive coefficient structures. Binary quartic, binary octic, τ-nonic, and ρ-nonic templates are pairwise incompatible.

These five theorems provide the architectural foundation for the perspective family program. They establish that construction constants are canonical (not arbitrary), algebraic (at rational P), degree-controlled (by the base field and polynomial), architecturally organized (binary, ternary, or mixed worlds), and structurally separated (by mutually exclusive coefficient normal forms). For pure prime-power degrees (2^k and 3^k), the world assignment is forced by the divisibility obstruction (Theorem D). For all exhibited families, the coefficient normal forms (Theorem E) provide an additional layer of separation.

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